

Air Quality Monitoring System

Minor Project Report — Embedded Systems

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1. Abstract

Air pollution is a growing concern in urban environments, contributing to respiratory illnesses and long-term health risks. This project presents a low-cost, real-time Air Quality Monitoring System built using an Arduino Uno microcontroller and an MQ135 gas sensor. The system continuously measures the concentration of harmful gases such as CO2, ammonia, and smoke in the surrounding air, displays the readings on a 16x2 LCD screen, and triggers an audible buzzer alert when pollution levels exceed a predefined safe threshold. The project demonstrates the practical application of embedded systems in environmental monitoring and can be extended with IoT connectivity for remote data logging and analysis.

2. Objective

To design and implement a real-time embedded system that monitors ambient air quality, provides immediate visual feedback through a display, and alerts users audibly when air quality deteriorates beyond safe limits.

3. Components Used

Component	Quantity	Function
Arduino Uno	1	Microcontroller — reads sensor data and controls outputs
MQ135 Gas Sensor	1	Detects air quality (CO2, NH3, smoke, etc.)
16x2 LCD (direct-wired)	1	Displays live air quality readings
10K Potentiometer	1	Controls LCD contrast
Buzzer	1	Sounds an alert when air quality is poor
Jumper Wires & Breadboard	As needed	Circuit connections

4. Circuit Connections

Component Pin	Arduino Pin
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MQ135 VCC	5V
MQ135 GND	GND
MQ135 AO (Analog Out)	A0
LCD RS	Digital Pin 7
LCD E	Digital Pin 6
LCD D4	Digital Pin 5
LCD D5	Digital Pin 4
LCD D6	Digital Pin 3
LCD D7	Digital Pin 2
LCD VSS	GND
LCD VDD	5V
LCD VO	Potentiometer wiper
LCD RW	GND
LCD A (backlight +)	5V
LCD K (backlight -)	GND
Buzzer +	Digital Pin 8
Buzzer -	GND

5. Working Principle

The MQ135 gas sensor continuously samples the surrounding air and produces an analog voltage proportional to the concentration of detected gases. This analog signal is read by the Arduino Uno's A0 pin and converted into a digital value (0–1023) using its built-in Analog-to-Digital Converter (ADC). This raw value is mapped to an approximate PPM (parts per million) scale to represent air quality in a human-readable format. The processed reading is displayed in real time on a 16x2 LCD screen wired directly to the Arduino's digital pins in 4-bit mode, with a potentiometer used to adjust display contrast. If the computed air quality value exceeds a predefined safe threshold, the Arduino activates a buzzer to alert nearby users of poor air quality, while simultaneously displaying a warning message on the LCD.

6. Arduino Code

The complete Arduino sketch (AirQualityMonitor.ino) is included in the project repository. Key logic:

```
LiquidCrystal lcd(7, 6, 5, 4, 3, 2);
int sensorValue = analogRead(mq135Pin);
int airQuality = map(sensorValue, 0, 1023, 0, 500);
lcd.setCursor(0, 1);
lcd.print(airQuality);
if (airQuality > threshold) {
  digitalWrite(buzzerPin, HIGH);
} else {
  digitalWrite(buzzerPin, LOW);
}
```

7. Result

The system successfully measures ambient air quality and displays real-time PPM values on the LCD screen. During testing, the buzzer reliably activated when exposed to smoke and gas sources, confirming correct threshold detection. The project validates the feasibility of using low-cost embedded components for practical environmental monitoring applications.

8. Future Scope

The system can be enhanced by integrating an ESP32 module to enable WiFi connectivity, allowing air quality data to be logged to a cloud platform (e.g., ThingSpeak or Firebase) for remote monitoring and historical trend analysis. Additional sensors (e.g., DHT11 for temperature/humidity, PM2.5 dust sensors) could be incorporated for a more comprehensive environmental assessment. A mobile app or web dashboard could also be developed for user-friendly visualization and alerts.

9. Conclusion

This project successfully demonstrates a functional, low-cost Air Quality Monitoring System using embedded systems principles. It provided practical exposure to sensor interfacing, ADC conversion, I2C communication, and real-time embedded programming — core skills applicable to further work in embedded systems and IoT development.